Chapter-5

Data Mining Architecture:

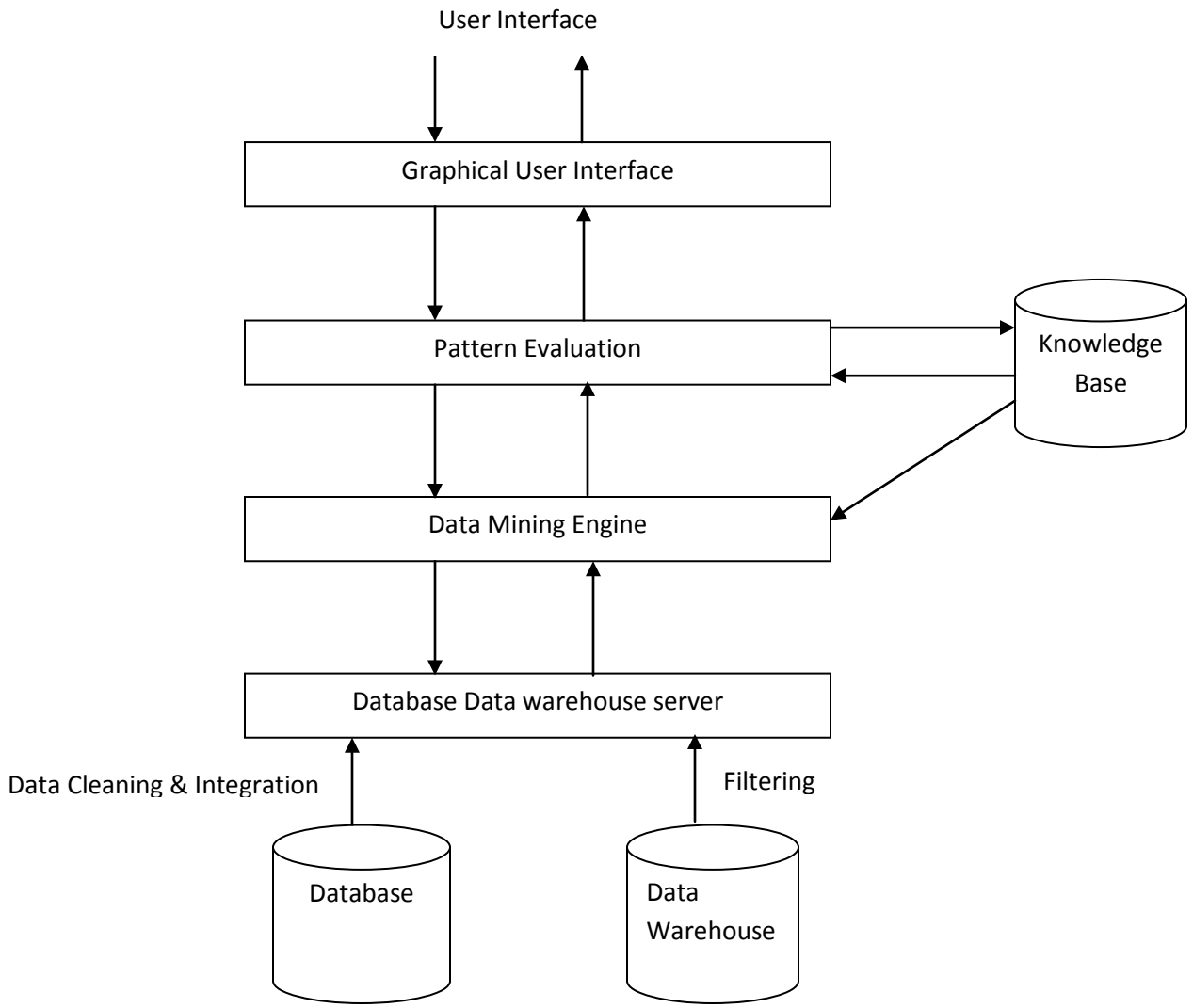


Figure: Data Mining Architecture

The data mining system architecture consists of following major components:-

[A] Database, Data warehouse or other information repository

This is one or a set of databases, data warehouses, spread sheets, or other kinds of information repositories. Data cleaning and data integration techniques may be performed on the data.

[B] Database or data warehouse server

The database or data warehouse server is responsible for fetching the relevant data, based on the user's data mining request.

[C] Knowledge Base

This is the domain knowledge that is used to guide the search, or evaluate the interestingness resulting patterns. Such knowledge can include concept hierarchies, used to organize attributes or attributes values into different levels of abstraction.

[D] Data Mining Engine

This is essential to the data mining system and ideally consists of a set of functional modules for tasks. Such as characterization, association, classification, clustering analysis and evaluation of deviation analysis.

[E] Pattern Evaluation Module

This component typically employs interestingness measures and interacts with the data mining modules. So as to focus the search towards interesting patterns. It may use interestingness thresholds to filter out discovered patterns.

[F] Graphical User Interface

This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structure, evaluate mined patterns, and visualize the patterns in different forms.

Data Mining Process:

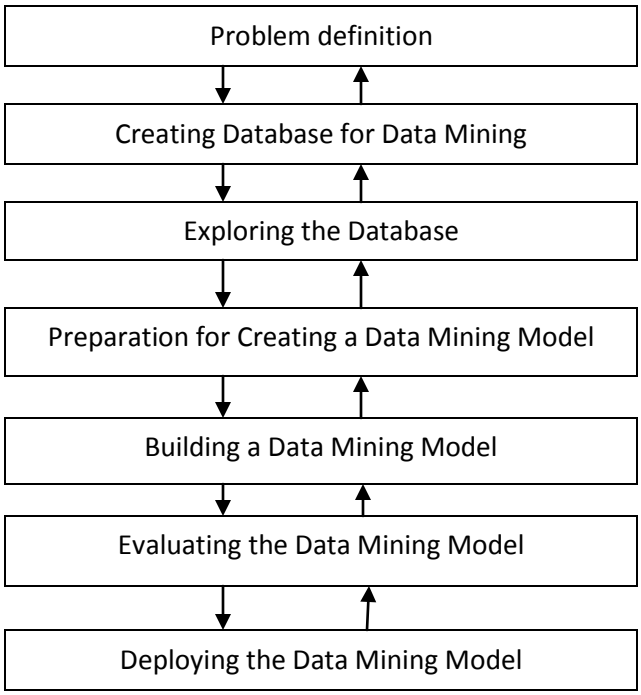


Fig: General phases of data mining process

A. **Problem definition :**
This phase is to understand the problem and the domain environment in which the problem occurs. You need to clearly define the problem before you proceed further problem definition specifies the limits within which the problem needs to be solved. It also specifies the cost limitation to solve the problem.

B. **Creating a database for data mining:**
This phase is to create a database where the data to be mined are stored for knowledge acquit ion. Creating a database does not require you to create specialized DBMS. You can even use a flat file or a spreadsheet to store data. The creation of data mining database consumes about 5o% to 90% of overall data mining process.

C. **Exploring the database:**
This phase is to select and examine important data sets of a data mining database in order to determine their flexibility to solve the problem. Exploring a database is a time consuming process. It requires a goal user interface and computer system with high processing speed.

D. **Preparation for creating a data mining model:**
This phase is to select various variables to act as predictors. New variables are also built depending upon the existing variables along with defining the range of variables in order to support the imprecise information.

E. **Building a data mining model:**
This phase is to create multiple data mining models and to select the best model. Building a data mining model is an iterative process. At times, you need to go back to the problem definition phase in order to change the problem definition itself. The data mining model that you select can be decision tree, neural network, or an association rule model.

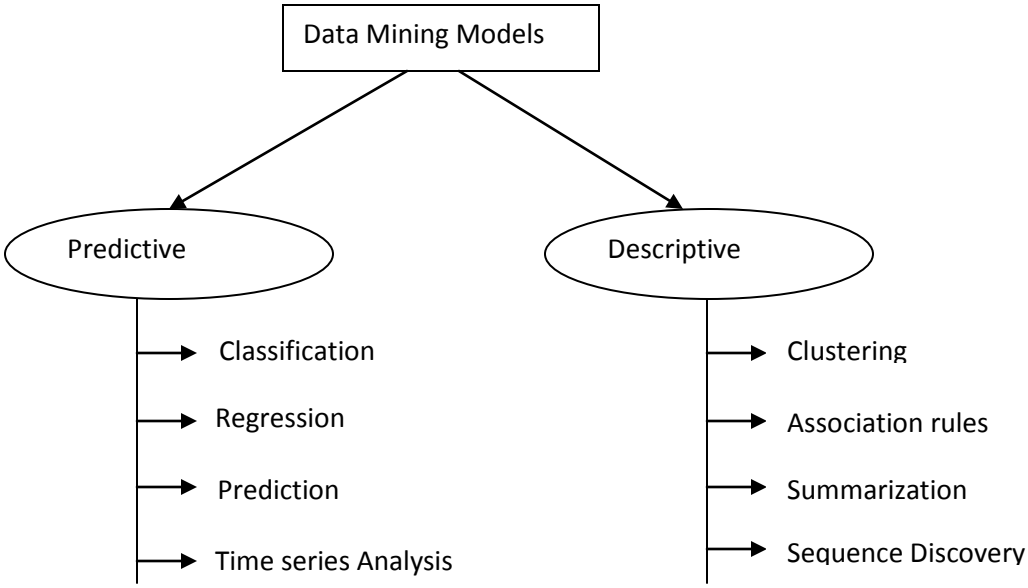
F. **Evaluating the data mining model:**
This phase is to evaluate the accuracy of the selected data mining model. In data mining, the evaluating parameter is data accuracy in order to test the working of the model. This is because the information generated in the simulated environment varies from the external environment. The error that occurs during the evaluation phase needs to be recorded and the cost/time involved in rectifying the error needs to be estimated. External validation is also needed to be performed in order to check whether the selected model performs correctly when provided real worlds values.

G. **Deploying the data mining model:**
This phase is to deploy the built and the evaluated data mining model in the external working environment. A monitoring system should monitor the working of the model and generate repots about its performance. The information in the report helps enhance the performance to selected data mining model.

Models of data mining

Data mining evolves many different algorithms to accomplish different tasks. All of these algorithm attempt to fit a model to the data. The algorithm attempt to fit a model to the data.

The algorithm examines the data and determines a model that is closest to the characteristics of the data being examined. There are two types of data mining models.



I. **Predictive Data models:**
It makes a prediction about values of data using known results found from different data. Dome of tasks of this model are classification, regression, prediction and time-series analysis.

II. **Descriptive data models:**
It identifies the patterns or relationships in data. Unlike the predictive model, a descriptive model serves as a way to explore the properties of the data examined not to predict new properties. Some of the tasks of this model are clustering, summarization, sequence discovery and association rules.

Data mining tasks and methodologies

1. **Classification:**
It maps the data into predefined group/classes. Patter recognition is its example where an input pattern is classified into one of several classes based on its similarity to these predefined classes.
2. **Regression:**
It is used to map the data item to a real valued prediction variable. It assumes that the target data fit into some known type of function and then determines the best function to model the data.
3. **Time series analysis:**
It examines the value of an attribute that varies over time. The values obtained are evenly spaced time (points/daily, weekly hours) etc

4. **Prediction:**
   It is a type of classification used for prediction the future data sets based on past and current data. E.g. Speech recognition, pattern recognition.
5. **Sequence Discovery:**
   It is used to analyze and discover the sequential patterns in data. These patterns are based on the time sequence of actions. E.g. most people who purchase the CD player may be found to purchase CDs within one week.
6. **Summarization:**
   It is also called characterization or generalization. It extracts or derives representative information about the database.
7. **Clustering:**
   It is similar to the classification except that the groups are not predefined, but rather defined by the data alone, clustering is alternatively referred to as unsupervised cleaning or segmentation. It is usually done by determining the similarity among the data on predefined attributes.
   Special type of clustering is called segmentation.
8. **Association rules:**
   Link analysis, alternatively referred to as affinity analysis or association, refers to the data mining task of uncovering relationship among data. An association rule is a model that identifies specific type of data associations. These associations are often used in retail sales community to identify items that are frequently purchased.

Data mining techniques:

Data mining techniques provides you with a way to use various data mining tasks. Such as: classification and clustering in order to predict solution for a problem. Some of the data mining techniques are:-

- Query tools
- Visualization techniques
- Decision tree
- Association rules
- Machine learning
- OLAP tools
- Statistical tools
- Neural networks
- Genetic algorithms
- Meta learning

I. **Visualization techniques**

Visualization refers to the visual representation of the data. This allows the users to summarize, extract and grasp the more complex results than more mathematical or text type description of the results e.g.

- Graphical: Pie chart, histograms, bar diagram, lines etc.
- Geometric: Box plot, scatter diagrams etc
- Icon based: Images, figures, colors etc
- Pixel based: Colored pixels
- Hybrid: Use of all above method in hybrid approach

II. **Statistical tools:**

Statistics is a branch of mathematics, which deals with the collection and analysis of numeric data by using various methods and techniques like:-

- Point estimation
- Data summarization
- Bayesian theorem
- Correlation
- Chi-square test (i.e. hypothesis test)
- Regression
- Mean, median, mode etc.

III. **Machine learning**

It is a sub discipline of computer science that deals with the design and implementation of learning algorithm. It is the process of generating a computer system that is capable independently acquiring the data and integrating that data to generate useful knowledge. The concept of machine learning is to make a computer system behave and act as a human being who learns from experience, analysis the observations made, such a system is developed that continuously self-improves and provides increased efficiency and effectiveness. The use of machine learning in data mining is that machine learning enables you to discover new and interesting structures and formats about a set of data that are previously unknown.

IV. **Decision tree:**

A decision tree is a predectine modeling technique used in classification, clustering and prediction task. Decision trees use a “divide and conquer” technique to split the problem search space into subsets. This represents a correct prediction.

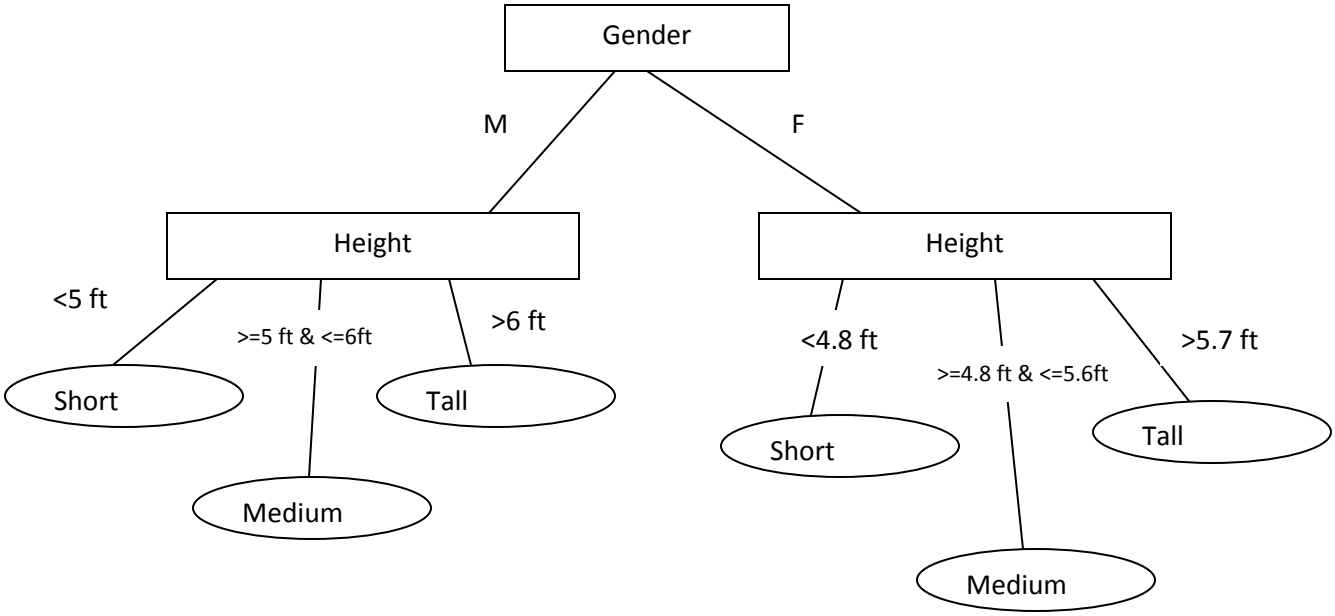
Definition:

A decision tree (DT) is a tree where the root and each internal node are labeled with a question. The arcs emanating from each node represents each possible answer to the associated question. Each leaf node represents a prediction of a solution to the problem under consideration.

A DT model is computational models that have 3-parts:-

a. A decision tree as defined above
b. An algorithm to create the tree
c. An algorithm that applies the tree to data and solves the problem under consideration.

Examples:



Algorithm:

Input:

T /* Decision tree*/

D /* Input database*/

Output:

M/*Model prediction*/

For each transaction in database Do

n=root node of T;

While n not leaf node do

Obtain answer to question on n applied to T;

Identify arc from T, which contains correct.

V. Neural Network (NN)

Neural network, often referred to as artificial neural network (ANN) are modeled after the working of human brain. The NN is actually an information processing system that consists of a graph representing the processing system as well as various algorithms that access that graph. As with the human brain, the NN consists of many connected processing elements. The NN then is structured as a directed graph with many nodes (processing elements) and arcs (interconnection) between them. The nodes in the graph are like individual neurons while arc are their interconnections.

The NN can be viewed as a directed graph with source (input), sink (output) and internal (hidden) nodes. The input nodes exist in an input layer, while the output node exists in an output layer. The hidden nodes exist over one or more hidden layers to perform a data mining task, a tuple is input through the input nodes and the output nodes determines what he prediction is?

A neural network model is a computational model consisting of 3 parts:-

a. Neural network graph that defines the data structure of the neural network
b. Learning algorithm that indicates how learning take place.
c. Recall technique that determines how information is obtained from the network.

Some of the forms of neural network are:-

- Perceptrons
- Back propagation network
- NN supervised learning
- Radial basis function network
- Kohonen-self organizing map

a) Perceptrons:

A Perceptron is a single neuron with multiple inputs and outputs. These are simple feed forward neural network. E.g.

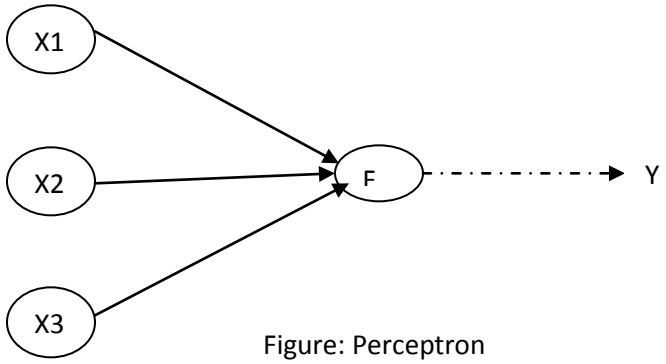


Figure: Perceptron

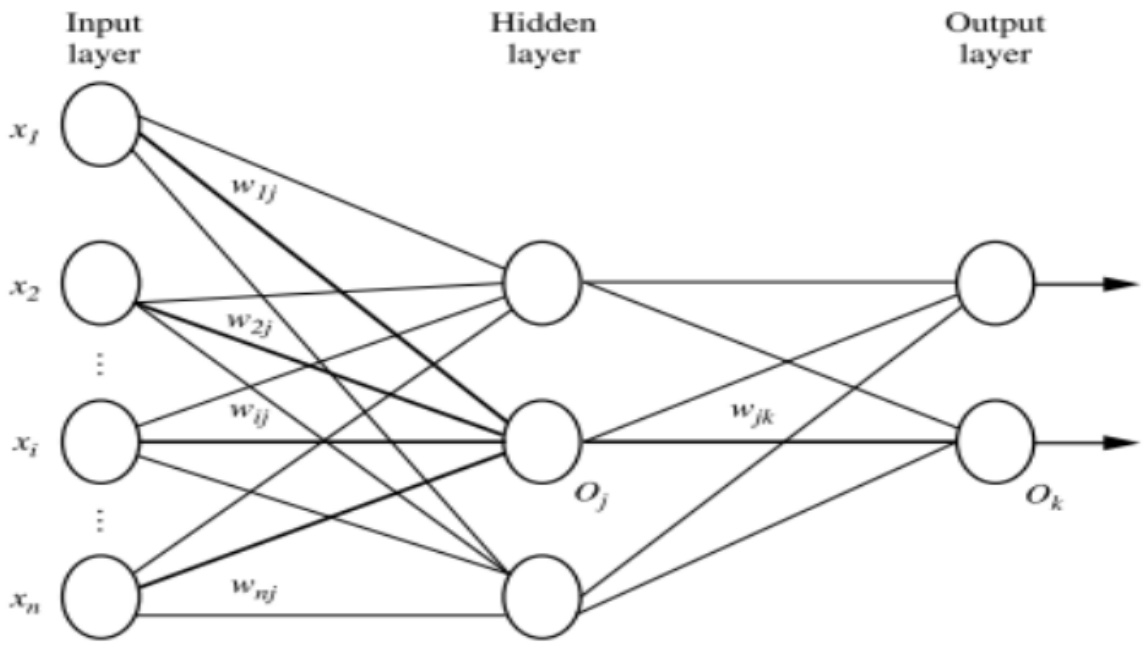
b) Back propagation network:

The back propagation algorithm performs learning on a multilayer feed-forward neural network. It iteratively learns a set of weights for prediction of the class label of tuples. A multilayer feed-forward neural network consists of an input layer, one or more hidden layers, and an output layer. An example of a multilayer feed-forward network is shown in figure below.

Each layer is made up of units. The inputs to the network correspond to the attributes measured for each training tuples. The inputs are fed simultaneously into the units making up the input layer. These inputs pass through the input layer and are then weighted and fed simultaneously to a second layer of "neuron like" units, known as a hidden layer. The outputs of the hidden layer units can be input to another hidden layer, and so on. The number of hidden layers is arbitrary, although in practice, usually only one is used. The weighted outputs of the last hidden layer are input to units making up the output layer, which emits the network's prediction for given tuples.

The units in the input layer are called input units. The units in the hidden layers and output layer are sometimes referred to as neurodes, due to their symbolic biological basis, or as output units.

Example:



c) Kohonen self-organizing map:

In 1981 Tuevo Kohonen demonstrated a completely different version of neural network called Kohonen self-organizing map. A Kohonen self-organizing map is a collection of neurons or units called its neighbors. It is a two dimensional map. Each node/unit contains a factor that is related to the space. In its initial setting, the map has a random

vectors assigned to each unit. Only at training stage, these vectors are adjusted to give a better coverage of the space.

VI. Genetic algorithm (GA):

Genetic algorithms are examples of evolutionary computing methods and are optimization type algorithms. GAs is directed search algorithm based on the mechanics of biological evolutions. It was first developed by John Holland in 1970s. it provides efficient, effective techniques for optimization and machine learning applications.

```
Algorithm
{
    Initialize population;
    Evaluate population
    While termination criteria not satisfied
    {
        Select parents for reproduction;
        Performs recombination and mutation;
        Evaluate population;
    }
}
```

VII. Meta learning:

It is the concept of combining the predictions made from multiple models of data mining and analyzing these predictions to formulate a new and previously unknown predictions. The concept of Meta learning is most useful when the models that you use vary widely and are difficult from each other.

Discovery and Analysis of patterns and trends

A data mining system has the potential to generate thousand or even millions of patterns or rules.

Pattern discovery:

The most common data mining technique used for mining the data is various types of traversal patterns. A traversal pattern is a set of pages visited by a user in a session. Other types of patterns may be uncovered by web usage mining. For example, association rules can look at pages accessed together in one session independent of ordering. Similar traversal patterns may be clustered together to provide a clustering of the users. This is different from clustering of pages, which tends to identify similar pages, not users. Several different types of traversal patterns have

been examined. These patterns differ in how the patterns are defined. The differences between the different types of patterns can be described by the following features.

- Duplicate page references may or may not be allowed.
- A pattern may be composed only of contiguous page references, or alternatively of any pages referenced in the same session.
- The pattern of references may or may not be maximal in the session. A frequent pattern is maximal if it has no sub pattern that is also frequent.

Pattern found using different combination of those three properties may be used to discover different features and thus may be used for different purposes.

Pattern Analysis:

Once patterns have been identified, they must be analyzed to determine how that information can be used. Some of the generated patterns may be deleted and determined not to be of interest.

Recent work has proposed examining web logs not only identify frequent types of traversal patterns, but also to identify patterns that are of interest because of their uniqueness or statistical properties. Patterns found need not have contiguous page reference.

Example:

Comparing the differences between traversal patterns of the customers an e-business site and those that are not-customers visitors to a site have been classified as short-time visitors, active investigations, and customers. Preprocessing first filters out the visitors who are short-time using concept hierarchies, the contents of the web pages are then obstructed to more general concept. The log is then divided into those for customers and those for non- customer. Each log is then examined to find patterns based on any desired requirements. The patterns found across the two logs are then compared for similarity.

Association analysis and cluster analysis:

Association Analysis:

Association analysis is the discovery of associarion rules showing attribute value condition that occurs frequently together in a given data set. Association rules are used when a data set has a large data items.

A typical example of association rule is market basket analysis. This process analyzes customer buying habits by finding association between the different items that customer place in their

shopping baskets. The discovery of such associations can help retailers develop marketing strategies by gaining insight into which items are frequently purchased together by customers.

Notation:

Terms	Description
D	database of transaction
Ti	transaction in D
@	Confidence
X, Y	item sets
X−>Y	Association rule
L	set of large item sets
l	large item set in L
C	set of candidate item sets
P	number of partitions

Support:

The support (S) for an association rule X=>Y is the percentage of transactions in the database that contains X∨Y.

Confidence:

The confidence or strength (X) for an association rule X=>Y is the ratio of the number of transactions that contains X∨Y to the number of transactions that contain X.

Definition:

Given a set of item I= {I1, I2…In} and database of transactions D = {T1, T2… Tn} where Ti= {Ii1, Ii2,…Iik} and Iij∈I, an association rule is an implication of the form X=>Y where X,Y, I are sets of item called item set and X∧Y=Φ.

Where X=>Y means “the database tuples that satisfy the conditions in X are likely to satisfy the condition in y”.

Example:

Simple data to illustrate association Rule.

Transactions	Computer	software
T1	√	-
T2	-	√
T3	√	√
T4	√	√
T5	√	-
T6	-	√
T7	-	√
T8	-	-
T9	-	-

Here, computer => software and software=> computer are two association rules. Find support (S) and confidence (@).

Case1: computer => software

$$\text{Support} = \frac{\{\text{Both}\}\ \text{count}}{\{\text{All transaction}\}\ \text{count}} *100\%$$

Therefore, Support= 2/7*100%

$$\text{Confidence} = \frac{\{\text{Both}\}\ \text{count}}{\{\text{Total computer}\}\ \text{count}} *100\%$$

Therefore, confidence= 2/4*100%

Case2:

Software => Computer

Similarly,

Support =2/7*100%

And Confidence =2/5*100%

Example:

{Contains (T, “Computer”) => Contains (T, “Software”)

[Support =1%, confidence =50%]

Which means that if a transaction T contain “Computer there is 50 % chance that it contains “Software” as well, and 1 % of all the transactions contain both.

The above rule can be written as follows

Computer => Software [1%, 50%]

Apriori Algorithm:

The Apriori algorithm is the most well known association rule algorithm and is used in most commercial products. The name of the algorithm is based on the fact that the algorithm uses prior knowledge of frequent item set properties.

Algorithm

Initialize K=1, C1= all the 1-itemset

Read the database to count the support of C1 to determine L1

L1= {frequent 1-itemset}

K==2 /* K represent the pass number */

While

($L_{k-1} \neq 0$) do

Begin

Ck = gen_candidate_itemsets with the given L_{k-1}

Prune (Ck)

For all transaction t∈T do

Increment the count of all candidates in Ck that are contained in T.

Lk= all candidates in Ck with minimum support

K:= K+1

End

Answer: Uk Lk

Or

The Apriori Algorithm

- Pseudo-code:

  C_k: Candidate itemset of size k

  L_k : frequent itemset of size k

  L_1 = {frequent items};

  for (k = 1; L_k !=∅; k++) **do begin**

  C_{k+1} = candidates generated from L_k;

  Important! How?! Next slide…

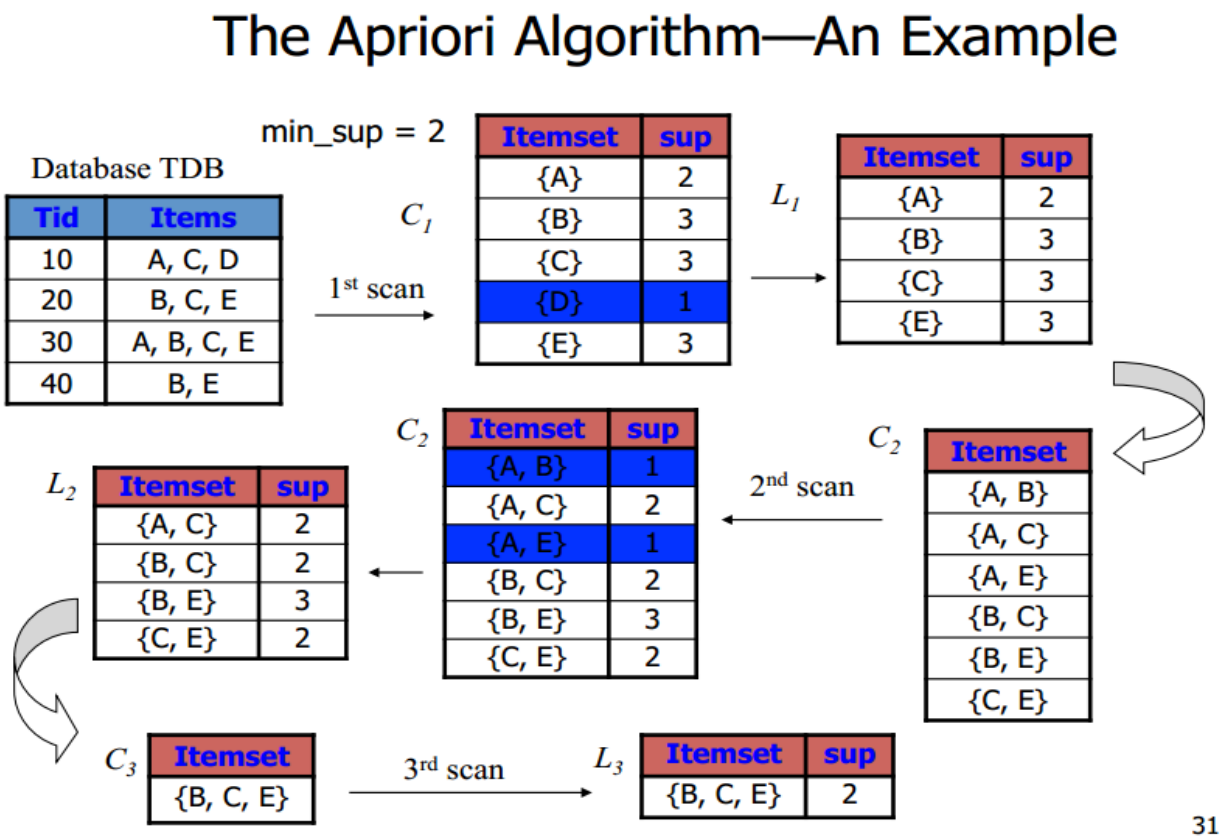
  for each transaction t in database do

  increment the count of all candidates in C_{k+1} that are contained in t

  L_{k+1} = candidates in C_{k+1} with min_support

  end

  return $\cup_k L_k$;



Other example:

Transactional data with support > 20%

T_id	**Item_set_id**
T1	I1, I2, I5
T2	I2, I4
T3	I2, I3
T4	I1, I2, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I1, I2, I3, I5
T9	I1, I2, I3

There are 9 transactions in the database i.e. |D| =9. We use the following figure to illustrate the Apriori algorithm for finding the frequent item sets in D.

Step 1:

Scan D for count of each candidate →

C1

Item set	Support count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

It scans all the transactions in order to count the number of occurrences of each item.

Step2:

It consists of candidate I-item sets satisfying minimum support i.e. L1 is determined.

Compare candidate support count with Minimum support count →

L1

Item set	Support count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

Step 3:

The algorithm uses L1 * L1 to generate candidate set of 2-item sets i.e. C2

Generate C2 candidate From L1 →

C2

Item sets
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

Step 4:

The transaction in D is scanned and the support count of each candidate item set in C2 is accumulated.

Scan D for count of each Candidate →

C2

Item sets	Support count
{I1, I2}	4
{I1, I3}	3
{I1, I4}	1
{I1, I5}	2
{I2, I3}	5
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

Step 5:

The set of frequent 2 item sets, L2 is then determined with minimum support of 2 from C2.

Compare candidate support count With minimum support count →

L2

Item sets	Support count
{I1, I2}	4
{I1, I3}	3
{I1, I5}	2
{I2, I3}	5
{I2, I4}	2
{I2, I5}	2

Step 6:

The generation of the set of candidate 3- item set C3 i.e. C3= L2 * L2 from L2

Generate C3 candidates From L2 →

C3

Item sets
{I1, I2, I3}
{I1, I2, I5}

Step 7:

The transaction in D are scanned in order to determine L3, consisting of candidate 3- item sets in C3 having minimum support.

Scan D for count Of each candidate →

C3

Item sets	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Step 8:

The algorithm uses L3 * L3 to generate a candidate set of 4- item sets, C4. But, C4=ϕ and the algorithm terminates by finding all frequent item sets.

Compare candidate support count

With minimum support count

Item sets	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Cluster Analysis:

A cluster analysis is the process of analyzing the various clusters to organize the different objects into meaningful and descriptive objects.

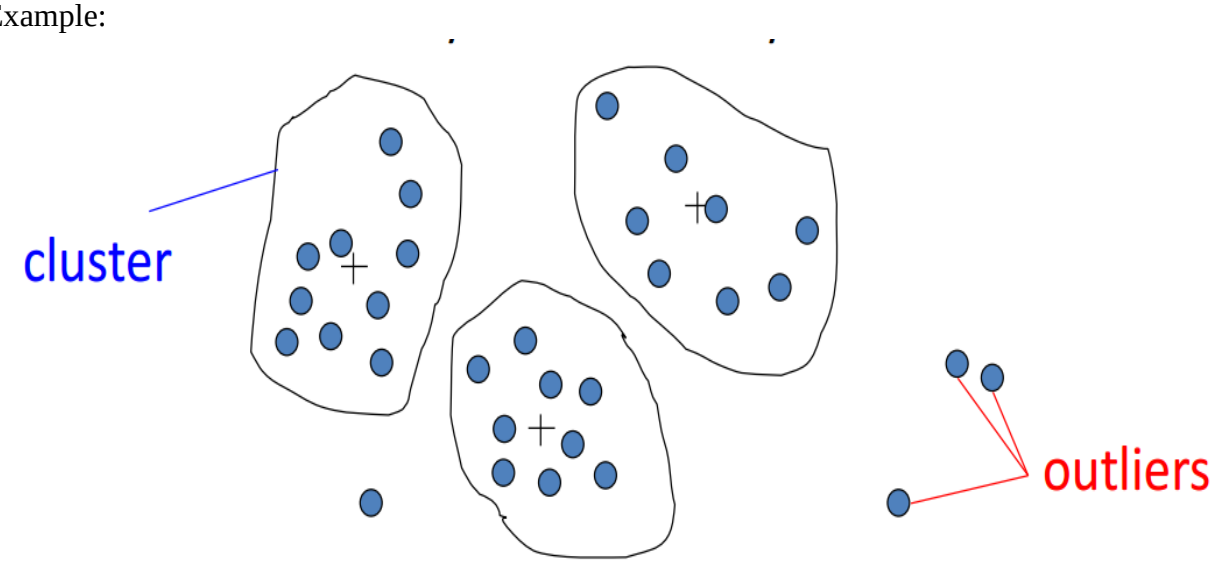
Cluster:

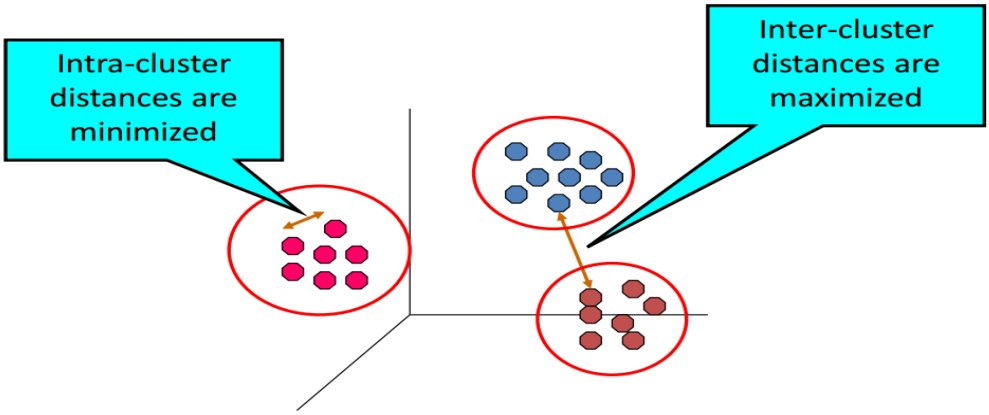
A cluster is a collection of data objects that are similar to one another within the same cluster and are dissimilar to the objects in other cluster.

Clustering:

It is a process of grouping the data into classes or clusters, so that objects within a cluster have high similarity in comparison to one another, but are very dissimilar to objects in other clusters. Thus, the process of grouping a set of physical or abstract objects into classes of similar objects is called clustering.

Example:





Euclidean distance measure between two data points:

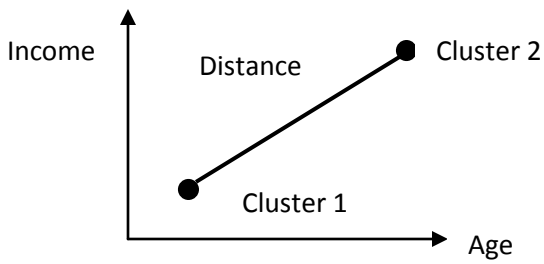


Figure: distance between two data points

Example 1:

	Age	Income	Credit
Customer 1	32	40,000	10,000
Customer 2	24	30,000	2,000
	8	10,000	8,000
		After rescaling	
	8	10	8

Now,

Distance = $\sqrt{(8^2+10^2+8^2)}$

$=15$

Note: we can use the following formula also

S.D = $\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$ =>for 2-D

S.D= $\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2+(Z_2-Z_1)^2}$ =>for 3-D

Example2

	Age	Income	Credit
Customer 1	80	15,000	1500
Customer 2	32	40,000	10,000
Customer 3	24	30,000	2,000

Taking customer 1 and customer 2

	Age	Income	Credit
Customer 1	80	15,000	1500
Customer 2	32	40,000	10,000
	48	-25,000	-8,500
		After rescaling	
	48	-25	-85

$S.D_{12} = \sqrt{((48)^2+(-25)^2+(-85)^2)}$

$= \sqrt{(3001.5)}$

=54.78

Now,

Taking C1 and C3

	Age	Income	Credit
Customer 1	80	15,000	1500
Customer 3	24	30,000	2,000
	56	-15,000	-500
		After rescaling	
	56	-15	-0.5

$$S.D_{13} = \sqrt{((56)^2 + (-15)^2 + (-0.5)^2)}$$

$$= \sqrt{(3261.25)}$$

$$=57.10$$

Now,

Taking C2 and C3

	Age	Income	Credit
Customer 1	32	40,000	10,000
Customer 2	24	30,000	2,000
	8	10,000	8,000
		After rescaling	
	8	10	8

$$S.D_{23} = \sqrt{((8)^2 + (10)^2 + (8)^2)}$$

$$= \sqrt{(64+100+64)}$$

$$=15.0996$$

Example 3:

Find the distance between the two data points

	Age	Yearly Income	Bank Balance
Data point 1	85	80,345	8,450
Data point 2	53	60,876	5,890

Taking data point 1 and data point 2

	Age	Income	Credit
Customer 1	85	80,345	8,450
Customer 2	53	60,876	5,890
	32	19,469	2,560
		After rescaling	
	32	19.469	2.56

$$S.D_{12} = \sqrt{((32)^2 + (19.469)^2 + (2.56)^2)}$$

$$= \sqrt{(1024+379.0419+6.5536)}$$

$$= \sqrt{(1409.5955)}$$

$$=37.54$$

Example 4:

Find the distance between the two data points

	Attributes1	Attributes 2	Attributes3
Data point 1	80	80,000	8,000
Data point 2	50	60,000	5,000

Taking data point 1 and data point 2

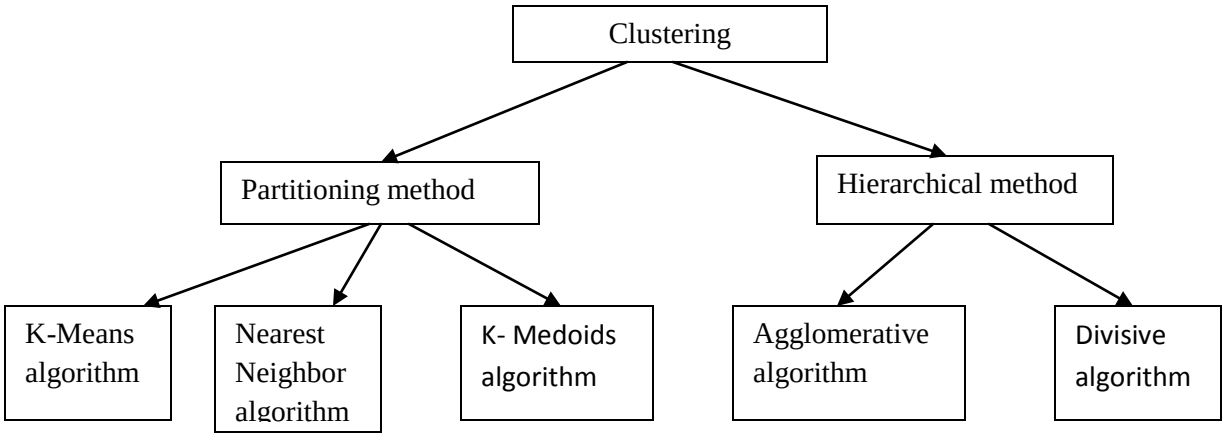
	Attributes1	Attributes2	Attributes3
Customer 1	80	80,000	8,000
Customer 2	50	60,000	5,000
	30	20,000	2,560
		After rescaling	
	30	20	3

$$S.D_{12} = \sqrt{((30)^2 + (20)^2 + (3)^2)}$$

$$= \sqrt{(900+400+9)}$$

$$= \sqrt{(949)}$$

$$=30.8058$$



1) Partitioning method:
   In this method, a partitioning algorithm arranges all the objects into various partitions, where total number of partitions is less then the total number of objects for examples, a database of n objects can be arranged into k partitions where k<n. Here each partition represents cluster. There are 3-types of partitioning algorithm:-
   - K-means algorithm
   - K-Medoids algorithm
   - Nearest Neighbor algorithm

2) Hierarchical method

A hierarchical clustering method works by grouping data objects into a tree of clusters. It is classified into two types, they are:

- Agglomerative hierarchical clustering
- Divisive hierarchical clustering

A. K- means algorithm

The k-means algorithm partitions all the n objects into k clusters, so that each cluster has maximum intra cluster similarity and minimum inter-cluster similarity. Cluster similarity is measured by the center of gravity of a cluster.

Example:

A database consists of 20 objects and user wants to arrange into 4 clusters. According to k-means algorithm, first randomly select 4-objects as the cluster center. Remaining objects are distributed to a cluster on the basis of cluster center to which the objects is similar. This distribution is enclosed by the dotted curves. The figure below shows the randomly selected objects.

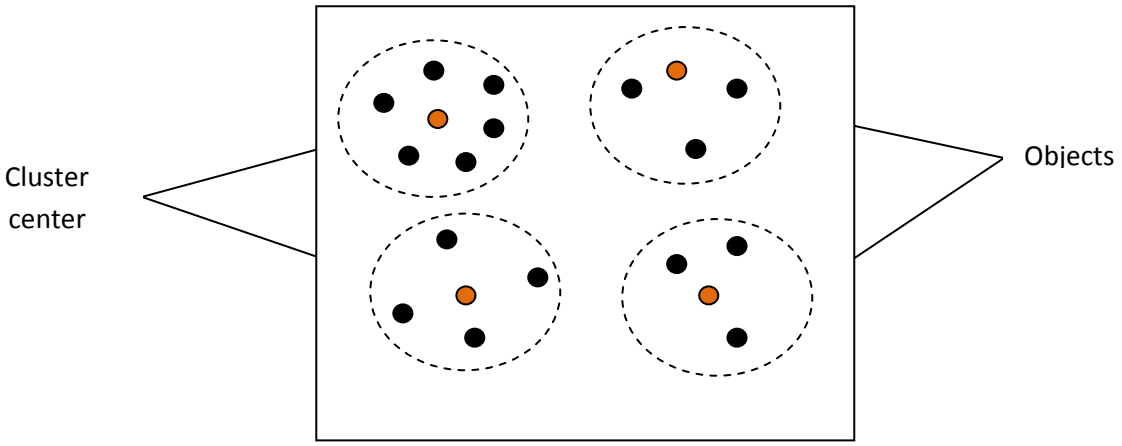


Figure: Randomly selected objects (i.e. K=4)

Now recalculate the mean of the clusters based on the objects that are presented in a cluster and redistribute the objects on the basis of new cluster center. This redistribution is also enclosed in dotted curves.

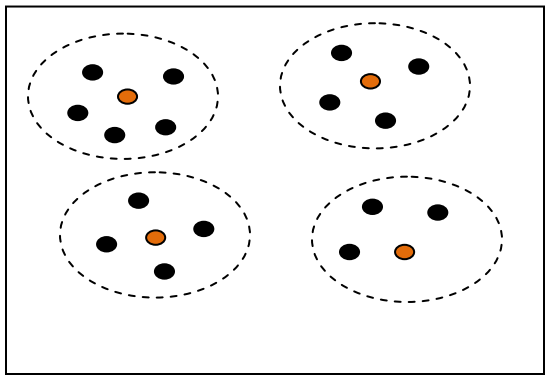


Fig: Redistribution of the objects

The re-clustering process is continued until redistribution is finished and the redistributing process returns the desired clusters. The resulting clusters are enclosed in solid curves.

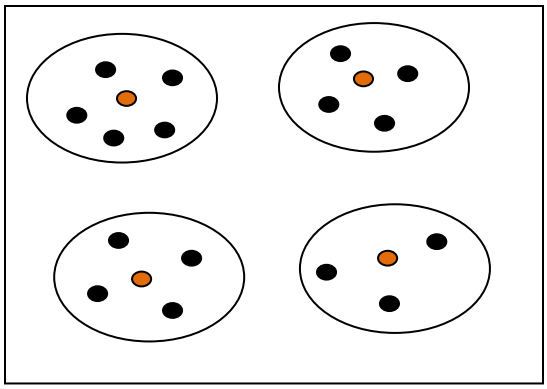


Fig: Final cluster

Algorithm:

Input:

D={t1,t2,t3,---,tn} //set of elements

K //number of desired clusters

Output:

K //set of clusters

The k-means Clustering Method(Algoritm)

Given k, the k-means algorithm is implemented in 4 steps:

1. Partition objects into k nonempty subsets.

2. Compute seed points as the centroids of the clusters of the current partition. The centroid is the center (mean point) of the cluster.

3. Assign each object to the cluster with the nearest seed point.

4. Go back to Step 2, stop when no more new assignment.

K-Means numerical example

2, 3, 6, 8, 9, 12, 15, 18, 22 – break into 3 clusters

Solution:

– Cluster 1 - 2, 8, 15 – mean = 8.3

– Cluster 2 - 3, 9, 18 – mean = 10

– Cluster 3 - 6, 12, 22 – mean = 13.3

• Re-assign

– Cluster 1 - 2, 3, 6, 8, 9 – mean = 5.6

– Cluster 2 – mean = 0

– Cluster 3 – 12, 15, 18, 22 – mean = 16.75

• Re-assign

– Cluster 1 – 3, 6, 8, 9 – mean = 6.5

– Cluster 2 – 2 – mean = 2

– Cluster 3 = 12, 15, 18, 22 – mean = 16.75

• Re-assign

– Cluster 1 - 6, 8, 9 – mean = 7.6

– Cluster 2 – 2, 3 – mean = 2.5

– Cluster 3 – 12, 15, 18, 22 – mean = 16.75

• Re-assign

– Cluster 1 - 6, 8, 9 – mean = 7.6

– Cluster 2 – 2, 3 - mean = 2.5

– Cluster 3 – 12, 15, 18, 22 – mean = 16.75

• No change, so we're done

B. Nearest Neighbor Algorithm

With this algorithm, items are iteratively merged into the existing clusters that are closet. In this algorithm a threshold t, is used to determine if items will be added to existing cluster or if a new cluster is created.

Algorithm:

Input:

D = {t1, t2, ---, tn} // set of elements

A //adjacency matrix

Output:

K // set of clusters

Nearest neighbor Algorithm:

K1 = {t1};

K = {K1};
K =1
For i=2 to n do
Find the tn in some cluster Km in K such that dis (ti, tm) is the smallest is dis (ti, tm) <=t then
Km=Km ∪ t
Else
K = K+1;
Kk = {ti};

Example

X1= Acid Durability (seconds)	X2= Strength (kg/square meter)	Y= Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with X1 = 3 and X2 = 7. Without another expensive survey, can we guess what the classification of this new tissue is?

1. Determine parameter K = number of nearest neighbors, assume k=3
2. Calculate the distance between the query-instance and all the training samples

Coordinate of query instance is (3, 7), instead of calculating the distance we compute square distance which is faster to calculate (without square root)

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)
7	7	$(7-3)^2+(7-7)^2=16$
7	4	$(7-3)^2+(4-7)^2=25$
3	4	$(3-3)^2+(4-7)^2=9$
1	4	$(1-3)^2+(4-7)^2=13$

3. *Sort the distance and determine nearest neighbors based on the K-th minimum distance*

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?
7	7	$(7-3)^2+(7-7)^2=16$	3	Yes
7	4	$(7-3)^2+(4-7)^2=25$	4	No
3	4	$(3-3)^2+(4-7)^2=9$	1	Yes
1	4	$(1-3)^2+(4-7)^2=13$	2	Yes

4. *Gather the category Y of the nearest neighbors.* Notice in the second row last column that the category of nearest neighbor (Y) is not included because the rank of this data is more than 3 (=K).

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?	Y = Category of nearest Neighbor
7	7	$(7-3)^2+(7-7)^2=16$	3	Yes	Bad
7	4	$(7-3)^2+(4-7)^2=25$	4	No	-
3	4	$(3-3)^2+(4-7)^2=9$	1	Yes	Good
1	4	$(1-3)^2+(4-7)^2=13$	2	Yes	Good

5. *Use simple majority of the category of nearest neighbors as the prediction value of the query instance*

We have 2 good and 1 bad, since 2>1 then we conclude that a new paper tissue that pass laboratory test with X1 = 3 and X2 = 7 is included in **Good** category.

C. K- Medoids Algorithm

It is also called the partitioning around Medoids (PAM) algorithm. This algorithm represents a cluster by a Medoids instead of center cluster. The partitioning is based on the sum of the dissimilarities between each objects and its reference point (Medoids).

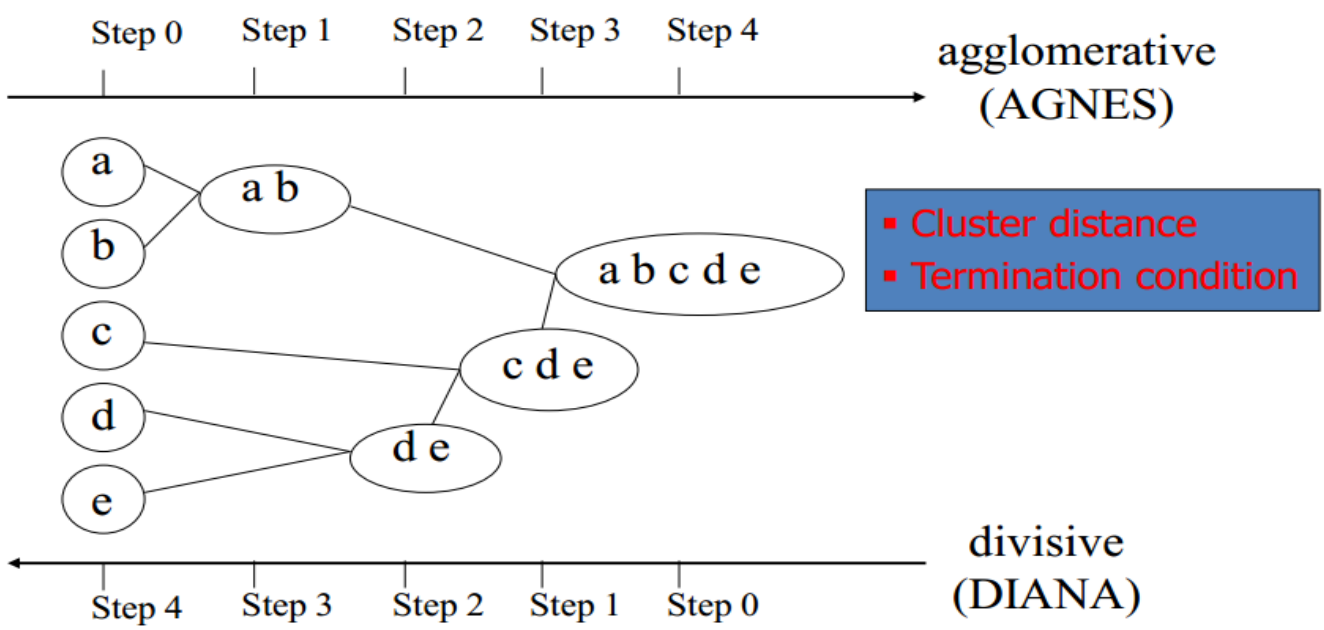
The K-Medoids algorithms works on following steps:

i. Randomly selected k- objects that represent a reference point, Medoids.
ii. Assign remaining objects to a cluster that is most similar to its Medoids.
iii. Randomly select a non- Medoids object orandom.
iv. Calculate the total cost C of swapping the medoid Oi with non-Medoids object orandom by cost function that is used to determine the dissimilarity between orandom and Oi.
v. Swap medoid Oi with the non-Medoids object orandom to make new Medoids, if total cost of swapping is negative.
vi. Process is again started from step 2 and it is repeated until no swapping occurs.

D. Agglomerative hierarchical clustering :

It works on the bottom-up approach. In this method, each object creates its own clusters. The single clusters are merged to make large cluster and the process of merging continues until all the singular clusters are merged into one big cluster that consists of all the objects.

E. Divisive hierarchical clustering

It works on the top-down approach. In this method all the objects are arranged with in a big single cluster and the large cluster is continuously divided into smaller clusters until each cluster has a single object.

Example: